

2025/05/24 Seminar

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§ 1 Recall

Theorem 1.1. *Suppose:*

(M^n, g) closed manifold with $n \geq 3$. (N^k, h) closed manifold; $\pi_l(N) \neq 0$ for some $l \geq 3$; admits no minimal 2-sphere.

Then $\exists$ nonconstant stationary harmonic map $u : M \rightarrow N$ s.t. $\dim(\text{Sing}(u)) \leq n - 3$ and $\text{Ind}_E(u) \leq l + 1$.

§ 2 Harmonic maps to spheres

In particular, taking $N = \mathbb{S}^k$ and $l = k$.

Combine with the work of Schoen and Uhlenbeck; Hong-Wang; Lin-Wang etc.

Corollary 2.1 (Cor. 1.3). *For any closed manifold M^n with $n \geq 3$ and for any $k \geq 3$, there is a nonconstant stationary harmonic map*

$$u_k : M \rightarrow \mathbb{S}^k$$

of Morse index $\text{Ind}_E(u) \leq k + 1$ and

$$\begin{cases} \dim(\text{Sing}(u)) \leq n - k - 1 & \text{if } 3 \leq k \leq 5 \\ \dim(\text{Sing}(u)) \leq n - 6 & \text{if } 6 \leq k \leq 9 \\ \dim(\text{Sing}(u)) \leq n - 7 & \text{if } k \geq 10 \end{cases}$$

In particular, if $3 \leq k \leq 5$, $k \geq n$, then u_k is smooth.

Proposition 2.2 (Prop. 1.4). *For any closed manifold (M^n, g) of dimension $3 \leq n \leq 5$, there exists $k_0(M, g) \in \mathbb{N}$ such that for any $k \geq k_0$ and any harmonic map $u : M \rightarrow \mathbb{S}^k$ with*

$$\text{ind}_E(u) \leq k + 1,$$

there exists a totally geodesic subsphere $\mathbb{S}^{k_0} \subset \mathbb{S}^k$ of dimension k_0 such that $u(M) \subset \mathbb{S}^{k_0}$.

§ 3 Applications to eigenvalue optimization

Motivation: Let Σ, g be closed Riemannian surface.

$$0 = \lambda_0(\Sigma, g) < \lambda_1(\Sigma, g) < \dots$$

the Laplace eigenvalues.

1. $\bar{\lambda}_m(\Sigma, g) := \lambda_m(\Sigma, g) \text{Vol}_g(\Sigma)$; normalized eigenvalue functionals
2. $\Lambda_m(\Sigma, [g]) = \sup_{h \in [g]} \bar{\lambda}_m(\Sigma, h)$; conformal class

According to Nadirashvili, and El Soufi-Ilias, metrics critical for $\bar{\lambda}_m(\Sigma, g)$ within the conformal class $[g]$ correspond to **harmonic** maps from $(\Sigma, [g])$ to spheres.

A natural way to generalize this to higher dimension:

$$\Lambda_m(M^n, [g]) := \sup_{h \in [g]} \bar{\lambda}_m(M^n, h) := \sup_{h \in [g]} \lambda_m(M^n, g) \text{Vol}(M^n, g)^{\frac{2}{n}}.$$

However, by Karpukhin, M., Metras, A., critical metrics for this problem correspond to n -harmonic maps. Recall the definition of p -harmonic map

$$E_g^p(\phi) := \frac{1}{p} \int_M |d\phi|_g^p dv_g,$$

which is conformally invariant when $p = n$.

Solution: Denote

$$0 = \lambda_0(\Sigma, g, \beta) < \lambda_1(\Sigma, g, \beta) < \dots$$

associated to

$$\Delta_g u = \lambda \beta u.$$

Note that $\lambda_m(\Sigma, e^{2\omega}g) = \lambda_m(\Sigma, g, e^{2\omega})$, so

$$\Lambda_m(\Sigma, [g]) = \sup_{0 < \beta \in C^\infty(\Sigma)} \left\{ \lambda_m(\Sigma, g, \beta) \int_\Sigma \beta dv_g \right\}$$

Proof here...

Definition 3.1. Introduce

$$\mathcal{V}_m(M^n, g) = \sup_{0 \leq \beta \in C^\infty(M^n)} \left\{ \lambda_m(M^n, g, \beta) \int_{M^n} \beta dv_g \right\}.$$

$\mathcal{V}_m(M^n, g)$ and $\Lambda_m(\Sigma, [g])$ share many similar property: $\Lambda_m(\Sigma, [g]) \leq Cm$; $\mathcal{V}_m(M^n, g) \leq Cm^{\frac{2}{n}}$.

Finally, and most importantly to us, we prove in Proposition 4.13 that densities critical for $\lambda_m(\beta) \|\beta\|_{L^1}$ are precisely the energy densities of classical sphere-valued harmonic maps $(M, g) \rightarrow \mathbb{S}^k$.

On a closed surface (Σ, g) , it is shown in [27](Karpukhin, M., Stern,) that, for the harmonic maps $u_k : (\Sigma, g) \rightarrow \mathbb{S}^k$ given by the two-dimensional analog of Corollary 1.3, the corresponding conformal metrics $g_k = \frac{1}{2}|du_k|_g^2 g$ maximize $\bar{\lambda}_1(\Sigma, g)$ in $[g]$ provided k is sufficiently large, i.e., $\Lambda_1(\Sigma, [g]) = \bar{\lambda}_1(\Sigma, g_k)$. The following theorem is a direct generalization of these results to higher dimensions ≤ 5 .

Theorem 3.2. *On any closed Riemannian manifold (M^n, g) of dimension $3 \leq n \leq 5$, for $k \geq k(M, g)$ sufficiently large, the energy densities $e(u_k) = |du_k|_g^2$ of the smooth harmonic maps u_k constructed in Corollary 1.3 realize $\mathcal{V}_1(M, g)$. Namely, the components of u_k are $\lambda_1(M^n, g, e(u_k))$ -eigenfunctions, $\lambda_1(M^n, g, e(u_k)) = 1$ and*

$$\mathcal{V}_1(M, g) = 2E(u_k).$$

§ 4 4

Proposition 4.1. *Let $u : (M, g) \rightarrow \mathbb{S}^k$ be a weakly harmonic map such that $\mu = |du|_g^2 dv_g$ is an admissible measure. Then u is smooth.*

Proposition 4.2. *For any $k \geq 3$ and any signed admissible measure ξ satisfying $v_2(\xi) \geq 0$, there exists a map $u \in W^{1,2}(M, \mathbb{S}^k)$ such that $E_g(u) \leq \mathcal{E}_k$ and*

$$\int u \varphi_0 dv_g = 0,$$

where φ_0 is a $v_1(\xi)$ -eigenfunction.

In particular,

$$\xi(M) \leq 2E_g(u) \leq 2\mathcal{E}_k, \tag{4.9}$$

with equality if and only if $v_2(\xi) = 0$, u is a smooth harmonic map of spectral index 1 and $\xi = |du|_g^2 dv_g$.

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